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**BCAAIMS 102**

**First Semester B.C.A. (AI&ML) Degree Examination,**
**December 2025/January 2026**
**(SEP) (2024 – 2025 Batch Onwards)**
**PROBLEM SOLVING USING C**

Time : 3 Hours Max. Marks : 80

***Note :*** *Answer **any ten** questions from Part – **A** and **one full** question from **each** Unit of Part – **B**.*

PART – A

1. a) What is Flowchart ? **(10×2=20)**
   b) What is pseudo code ?
   c) List rules to be followed while naming an identifier.
   d) Differentiate pre and post increment operators in C.
   e) Differentiate break and continue statements.
   f) Write general form of goto statement. Give an example.
   g) Specify how to initialize two dimensional arrays ? Give example.
   h) What is nesting of functions ? Give an example.
   i) Specify how gets( ) function is more useful compare to scanf( ) function with reference to reading of string value.
   j) What is recursion ?
   k) What are actual and formal arguments ?
   l) List any two advantages of pointers.

PART – B

**Unit – I**

2. a) With the help of diagram explain different phases of Program Development Cycle.
   b) List and explain any five symbols of Flowchart.
   c) What are constants ? How they are classified ? Give example for each. **(5+5+5)**

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3. a) Explain Algorithm with its properties. Give an example.
   b) List and explain any five features of C.
   c) Explain primary data types available in C. **(5+5+5)**

**Unit – II**

4. a) Explain scanf() function with syntax and example.
   b) What is type conversion ? Explain different types of type conversion.
   c) Explain switch statement with syntax and example. **(5+5+5)**

5. a) List and explain arithmetic and relational operators in C with examples.
   b) Explain different forms of if statement with syntax and examples.
   c) Explain printf() function with syntax and example. **(5+5+5)**

**Unit – III**

6. a) How do you declare, initialize and access a one dimensional array ?
   b) Explain with example declaring, defining and calling a function.
   c) Explain do..while statement with syntax and example. **(5+5+5)**

7. a) List and explain any five string handling functions with examples for each.
   b) Explain with example any two categories of user-defined functions.
   c) Explain automatic and static variables with example. **(5+5+5)**

**Unit – IV**

8 a) What is file ? Explain any five file access modes.
   b) Explain structure with syntax and example.
   c) List and explain any five file handling functions used for file operations. **(5+5+5)**

9. a) What is a pointer ? How to declare and initialize pointer variable ? Give examples.
   b) Explain arrays of structures with the help of an example.
   c) What is union ? How it is different from structure ? Explain with example. **(5+5+5)**